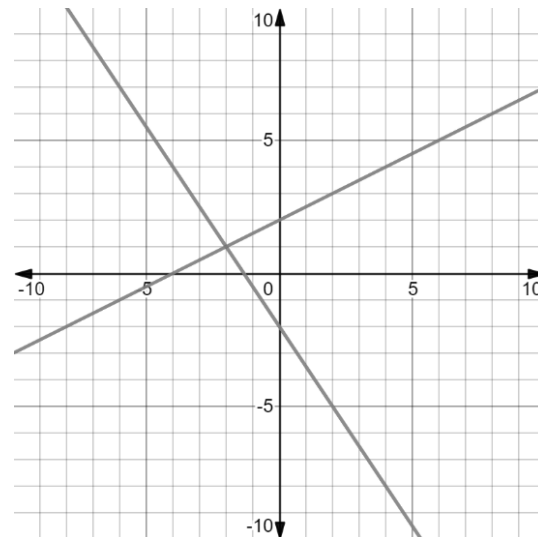


Algebra 1
Unit 5
Lesson 6 Practice

Name _____
Date _____
Period _____

The graph of the system of equations $\begin{cases} x - 2y = -4 \\ -3x - 2y = 4 \end{cases}$ is shown.



1. To determine the solution to a system of equations, identify the
- ☐ point of intersection.
 - ☐ slope.
 - ☐ y-intercept.

2. Determine the solution to the system of equations from the graph.

Verify algebraically that your solution satisfies both equations in the system.

3. $x - 2y = -4$

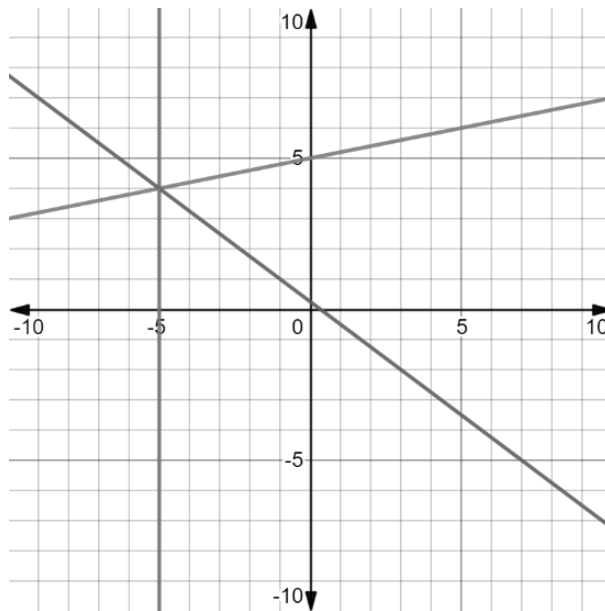
4. $-3x - 2y = 4$

5. Choose all of the systems of equations that are equivalent to the original system of equations, $\begin{cases} x - 2y = -4 \\ -3x - 2y = 4 \end{cases}$. Explain how you know the systems are equivalent.

$\begin{cases} y = \frac{1}{2}x + 2 \\ y = -\frac{3}{2}x - 2 \end{cases}$	$\begin{cases} y = -\frac{1}{2}x + 2 \\ y = -\frac{3}{2}x - 2 \end{cases}$	$\begin{cases} 4x - 8y = -16 \\ -3x - 2y = 4 \end{cases}$	$\begin{cases} x - 2y = -4 \\ -x - \frac{2}{3}y = \frac{4}{3} \end{cases}$
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6. Verify your choices in question 5 algebraically.

The graphs of the system of equations $\begin{cases} y = \frac{1}{5}x + 5 \\ y = -\frac{3}{4}x + \frac{1}{4} \end{cases}$ and $\begin{cases} x = -5 \\ y = \frac{1}{5}x + 5 \end{cases}$ are shown.



7. The solution to the system $\begin{cases} y = \frac{1}{5}x + 5 \\ y = -\frac{3}{4}x + \frac{1}{4} \end{cases}$ is (____, ____).

8. The solution to the system $\begin{cases} x = -5 \\ y = \frac{1}{5}x + 5 \end{cases}$ is (____, ____).

9. Fill in the blanks with words from the word bank to complete the sentence.

The two systems of equations are _____ because the two systems result in _____ solution(s).	Word Bank: ☐ equivalent ☐ equal ☐ not equivalent ☐ not equal ☐ the same ☐ different
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10. Write a third system of equations that is equivalent to the system, $\begin{cases} y = \frac{1}{5}x + 5 \\ y = -\frac{3}{4}x + \frac{1}{4} \end{cases}$.